FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 2HA-I 24004/24005 — 40043 (Version B) / 42042 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

PHYSICS HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 40043 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. If the velocity of free electron is doubled, its De-Broglie wave length changes by the factor of:

(A) 2 (B) 1/2

(C) 1/√2 (D) √2

Q2. The energy of photon of wave length 500nm is:

(A) 1.77 ev (B) 1.52 ev

(C) 3.10 ev (D) 2.49 ev

Q3. An expression for current gain of a transistor is given by:

(A) $\beta = I_E / I_B$ (B) $\beta = I_B / I_E$

(C) $\beta = I_B / I_C$ (D) $\beta = I_C / I_B$

Q4. For an atom in excited state n=5, maximum number of transitions that take place is:

(A) 10
(B) 25
(C) 3
(D) 5

Q5. The variation of electric potential due to a point charge with distance is represented by the graph:

(A) (A constant ΔV vs Δr graph is printed in the source scan.)
(B) (A rising-curve ΔV vs Δr graph is printed in the source scan.)
(C) (A straight-line ΔV vs Δr graph is printed in the source scan.)
(D) (A decreasing-line ΔV vs Δr graph is printed in the source scan.)

Q6. In RLC series circuit, the current at resonance frequency is:

(A) Zero
(B) Infinite
(C) Maximum
(D) Minimum

Q7. In the figure, the charge on 3 μF capacitor is: (A circuit diagram of 2 μF, 3 μF and 6 μF capacitors in series with V = 20 V is printed in the source scan.)

(A) 30 μC
(B) 40 μC
(C) 10 μC
(D) 20 μC

Q8. The force acting on a charged particle kept between the plates of charged capacitor is F. If one of the plates of the capacitor is removed, the force on particle becomes:

(A) F/2
(B) Zero
(C) F
(D) 2F

Q9. The best instrument for the accurate measurement of emf of a cell is:

(A) Potentiometer
(B) Ohm meter
(C) Voltmeter
(D) Ammeter

Q10. The AC circuit in which current and voltage are in phase, the power factor is:

(A) -1
(B) 2
(C) Zero
(D) 1

Q11. The temperature below which the resistivity of super conductor falls to zero is called:

(A) Critical temperature
(B) Kelvin temperature
(C) Absolute temperature
(D) Limiting temperature

Q12. A 60 watt bulb operates at 50% efficiency, the energy dissipated in 10 Seconds is:

(A) 600 J
(B) 700 J
(C) 200 J
(D) 300 J

Q13. Galvanometer is sensitive when the value of C/BAN is:

(A) Zero (B) Negative

(C) Small (D) Large

Q14. With the increase in speed of motor, the magnitude of back emf:

(A) Decreases (B) Becomes zero

(C) Does not change (D) Increases

Q15. If N_s/N_p = 1 then the transformer is:

(A) Neither step up nor step down (B) Rectifier transformer

(C) Step up (D) Step down

Q16. A wire is stretched to double of its length its strain is:

(A) 0.5 (B) 0

(C) 2 (D) 1

Q17. PN junction when reverse biased acts as:

(A) Inductor (B) OFF switch

(C) Capacitor (D) ON switch

PHYSICS HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 24005 (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14 x 3 = 42)

درج سوالات کے مختصر جوابات تحریر کریں۔

(i) [03] What is meant by 'Magnetic Flux' and 'Magnetic Flux density'? Also give their units. — OR (03) — Why energy dissipated per cycle (magnetization and demagnetization). For steel is more as compared to iron? Explain briefly.

(ii) [1+2] What is Wein's displacement law? Explain briefly. — OR (03) — Prove that E = -ΔV/Δr

(iii) [03] Differentiate between controlled and uncontrolled nuclear chain reactions. How is the chain reaction controlled? — OR (03) — Why in a transistor the base region is made thin and lightly doped?

(iv) [03] Distinguish between N-type semiconductor and P-type semiconductor. — OR (03) — Calculate the longest wave length of radiation for the Lyman series of hydrogen spectra.

(v) [03] Differentiate "Curie temperature" and "Critical temperature". — OR (2+1) — How capacitor discharge ignition system works? Enlist some of its uses.

(vi) [1+2] Briefly explain resistivity. How it depends on temperature? — OR (2+1) — How can a galvanometer be converted into voltmeter? Draw circuit diagram as well.

(vii) [03] How are eddy currents produced? Identify their heating effects. — OR (03) — How Geiger-Muller counter detects and counts radiation?

(viii) [03] In transformer, why laminated iron core is used instead of solid one? — OR (03) — A coil having a resistance of 10 Ohm and an inductance of 32 mH is connected to 220 V, 50 Hz AC supply. Calculate current passing through the coil.

(ix) [03] What is meant by peak value and effective value of sinusoidal current? Give relation between them. — OR (03) — Differentiate Paramagnetic and Diamagnetic materials with one example each.

(x) [03] Briefly explain working of transistor as a switch. — OR (03) — What is meant by Meta-stable state and population inversion for LASER action?

(xi) [03] Calculate De-Broglie wave length of an electron having KE = 1200 kev. — OR (03) — Discuss the difference between Hadrons and Leptons.

(xii) [03] How much energy is released when 0.5 kg of U-235 undergoes fission reaction? (If the disintegration energy per event is Q = 208 Mev.) — OR (03) — What is meant by Alpha factor and Beta factor for common emitter configuration of transistor? Derive relation between them.

(xiii) [2+1] Briefly explain the principle of metal detector with circuit diagram. — OR (1+2) — What is electron volt (ev)? Derive its relation with SI unit of energy.

(xiv) [03] Under what condition a source (Battery or cell) gives maximum power output? Discuss briefly. — OR (1+2) — State Lenz's law. Prove that it is according to law of conservation of energy.

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 3+4) Q.3 State postulates of Bohr's atomic model. Show that energy of the electron in H-atom is quantized. — OR (5+2) — Derive an expression for charge to mass ratio for an electron and then calculate its value.

Q4. (Marks 1+2+4) Q.4 What is photoelectric effect? Why classical physics fails to explain photoelectric effect? Derive Einstein photoelectric equation. — OR (1+5+1) — What is potentiometer? Explain its principle, construction and working. Also give some uses.

Q5. (Marks 3+3) Q.5 State and explain Gauss's law. Find electric field intensity between two oppositely charged parallel plates. — OR (3+1+2) — Explain the phenomenon of self-inductance of a coil. What is its unit? On what factors self-induction depends?

Q6. (Marks 2+1+3) Q.6 What is RLC series resonance circuit? Draw its impedance diagram. Also give its properties. — OR (2+4) — What is meant by 'half-life' of a radio-active element? Show that $T_{1/2} = 0.693/\lambda$

VERSION D

PHYSICS HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 42042 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. What is the SI unit of current gain?

(A) It has no unit
(B) Ampere
(C) Volt
(D) Ohm-meter

Q2. The variation of electric potential due to a point charge with distance is represented by the graph as:

(A) (A constant ΔV vs Δr graph is printed in the source scan.)
(B) (A straight-line ΔV vs Δr graph is printed in the source scan.)
(C) (A decreasing-line ΔV vs Δr graph is printed in the source scan.)
(D) (A rising-curve ΔV vs Δr graph is printed in the source scan.)

Q3. The interior of a hollow charged sphere is a:

(A) Field free region
(B) Strong field region
(C) Weak field region
(D) Partial field region

Q4. The thermocouple circuit which is used to measure temperature, works on:

(A) Compton effect
(B) Thomson effect
(C) Peltier effect
(D) Seebeck effect

Q5. Which is correct for an open circuit?

(A) V_t = 0
(B) V_t < ε
(C) V_t = ε
(D) V_t = ε + Ir

Q6. What fraction of a radio-active sample will be left after 600 years whose half-life is 150 years?

(A) 1/16
(B) 1/2
(C) 1/4
(D) 1/8

Q7. Two up quarks and one down quark make:

(A) A Meson
(B) A Proton
(C) A Neutron
(D) A Photon

Q8. When an electron in an atom goes from a lower orbit to higher orbit its:

(A) KE and PE decrease
(B) KE increases, PE decreases
(C) KE decreases, PE increases
(D) KE and PE increase

Q9. An electron moves 3×10^5 m/s perpendicular to magnetic field of 2 T. What is the magnitude of magnetic force?

(A) 4.0×10^6 N
(B) 9.6×10^-17 N
(C) 6.4×10^-17 N
(D) 3.6×10^-10 N

Q10. If the length of solenoid is doubled but number of turns remain same, then B inside the solenoid becomes:

(A) Four times
(B) Double
(C) One fourth
(D) Half

Q11. The term Δϕ/Δt has the same dimensions as:

(A) Current
(B) Flux
(C) Potential difference
(D) Time

Q12. If back emf is zero, then the coil draws:

(A) Minimum current
(B) Steady current
(C) Zero current
(D) Maximum current

Q13. In the equation Δλ = h/(m_o c) (1 - Cosθ), which factor is called Compton wave length?

(A) h(1 - Cosθ)
(B) h / m_o c
(C) 1 / m_o c
(D) 1 - Cosθ

Q14. In choke coil the reactance X_L and resistance R_s are related as:

(A) X_L = ∞
(B) X_L = R
(C) X_L << R
(D) X_L >> R

Q15. In purely inductive circuit, the current:

(A) Lags the emf by 60°
(B) Lags the emf by 90°
(C) Leads the emf by 90°
(D) Is in phase with emf

Q16. Impurity atoms are doped in semi-conductor to increase:

(A) Resistivity
(B) Protons
(C) Positrons
(D) Conductivity

Q17. Curie point is a temperature where:

(A) Paramagnetic changes to Ferromagnetic material
(B) Diamagnetic changes to Paramagnetic material
(C) Paramagnetic changes to Diamagnetic material
(D) Ferromagnetic changes to Paramagnetic material

PHYSICS HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 24005 (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14 x 3 = 42)

درج سوالات کے مختصر جوابات تحریر کریں۔

(i) [03] Water has a large dielectric constant, but it is rarely used in capacitors. Why? — OR (2+1) — What is meant by moderator, used in atomic reactors? Give example.

(ii) [03] How will capacitance of a parallel plate capacitor be affected if area of plates is doubled and separation between them is halved? — OR (03) — What is meant by 'critical mass', 'sub-critical mass' and 'super critical mass' for fission chain reaction?

(iii) [03] How does stator help rotator to rotate in A.C. motor? — OR (03) — As rate of doing work increases by motor, the back emf decreases. Why?

(iv) [03] How can a rheostat be used as potential divider? Draw circuit diagram as well. — OR (03) — What is meant by alpha factor and beta factor for common emitter configuration of transistor? Give relation between them.

(v) [03] Calculate the current flowing through a circuit of resistance 1500 ohm connected with a battery of emf 100 V with internal resistance 0.01 Ohm. — OR (03) — Briefly explain the terms 'magnetic flux' and 'magnetic flux density'? Give their units as well.

(vi) [1+2] Can an electron at rest be set in motion with a magnet? Explain briefly. — OR (03) — Differentiate between spontaneous and stimulated emissions. Also show in diagrams.

(vii) [03] What is the time period of an electron projected into a uniform magnetic field of 20 mT and moves in a circle of radius 6 cm? — OR (03) — Why is common emitter configuration of transistor widely used in amplifier circuits?

(viii) [1+2] What is choke coil? Give its importance in A.C. circuits. — OR (03) — How are eddy currents produced? Identify their heating effects.

(ix) [03] What determines the gradient of a graph of inductive reactance against frequency? Explain briefly. — OR (2+1) — Write Stefan-Boltzmann law for black body radiation. Give its mathematical expression as well.

(x) [03] Differentiate, between 'critical temperature' and 'curie temperature' with examples. — OR (03) — How a galvanometer can be converted into ammeter? Explain with the help of diagram also derive formula.

(xi) [03] Why in a transistor, the base is thin and lightly doped? Explain briefly. — OR (03) — A 24.0 V car battery powers a 30.0 watt bulb. How many charges pass through it, in each second?

(xii) [03] Which factors cause to produce magnetic field in an atom? Explain briefly. — OR (03) — How did De-Broglie prove the third postulate of Bohr's atomic model? Explain briefly.

(xiii) [1+2] Under what condition Compton shift has maximum wavelength? Also calculate Compton shift wavelength. — OR (2+1) — What is meant by depletion layer in PN-Junction? How is it developed? Explain briefly.

(xiv) [03] When a solid is heated it begins to glow, why does it first appear red? — OR (1+2) — How many basic forces of nature exist? Describe any two briefly.

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 1+6) Q.3 Explain the concept of electric potential. Derive an expression for electric potential at a point in electric field due to a point charge. — OR (3+4) — State postulates of Bohr's atomic model. Show that radii of the orbit of H-atom are quantized.

Q4. (Marks 3+3+1) Q.4 What is meant by emf (ε), internal resistance (r) and terminal potential difference (Vt) of a battery? Derive relation between them. Under what condition Vt > ε? — OR (1+2+4) — What is AC generator? How is an AC generator used to produce an alternating current? Derive mathematical expression of A.C. Also show it graphically.

Q5. (Marks 2+2+2) Q.5 The Half life of Radium is 5.0×10^10 S. A sample contains 6.0×10^16 nuclei. Calculate. a. Decay constant. b. How many radium nuclei will decay per second? Express your answer in curies. — OR (06) — A sinusoidal alternating voltage of angular frequency ω is connected across a capacitor C. Find mathematical expression for instantaneous voltage, current and average power dissipated per cycle of applied voltage.

Q6. (Marks 06) Q.6 Explain the phenomena of pair production and pair annihilation. — OR (06) — Describe the magnetic properties of material explained on the basis of B-H curve.